

CH5440 MULTIVARIATE DATA ANALYSIS FOR PROCESS MODELING

Midterm Exam

Instructions:

- a) **Duration:** 90 Minutes
- b) Answer **ALL** Questions
- c) Two Sided (2 Pages) A4 Formulae Sheet Permitted
- d) You may work out the answers to your questions using a scribble pad (need not be submitted after exam)
- e) Additional electronic devices like smart watches, etc. are NOT allowed.
- f) Use only the online scientific calculator
- g) **Total Marks:** 45
- h) **Multiple Choice Type:** - 0.5 marks penalty for wrong answer (applicable to all sections)
- i) **Numerical Answer Type (NAT) Questions:** No negative marking for wrong answers
- j) **Pick ONE Correct Choice in MCQ.**

SECTION A

All Questions in this section carry ONE mark each.

1. Fill in the blanks by choosing the correct options: Tom Peters in his book on “Thriving on Chaos” has said “We are _____ in information and starved for _____
(A) drowning, data
(B) rich, data
(C) drowning, knowledge
(D) lacking, knowledge
2. Businesses use the collected data intelligently to garner actionable insights for
(A) Improved decision-making
(B) Predictive modeling

(C) Accumulate business intelligence

(D) All the above

3. Sensors are fixed on equipment to collect data. Patterns are then identified and predictions may be made when the equipment might malfunction. This is called **Predictive Maintenance**

4. In a chemical plant, the tons produced of a valuable commodity is expressed in Table 1 below.

Table 1. Production Data (Question 4)

Reactor	January Average Production (tons/day)	February Average Production (tons/day)	March Average Production (tons/day)
A	12	16	14
B	10	11	14
C	14	12	16

The above information is an example of

(A) Only time series data

(B) Only cross-sectional data

(C) Both time series and cross-sectional data

(D) Neither time series nor cross-sectional data

5. Apartment numbers such as 12, 14, 15, 16 are examples of **Nominal** variables.

6. Pick the correct choice

(A) Both gauge pressure and absolute pressure are interval variables

(B) Both gauge pressure and absolute pressure are ratio variables

(C) Gauge pressure is ratio variable, but absolute pressure is interval variable

(D) Gauge pressure is interval variable, but absolute pressure is ratio variable

7. State TRUE or FALSE

Lieutenant, Captain, Major, and Colonel are examples of NOMINAL variables.

FALSE.

8. A categorical variable has 4 levels. Each level is represented by 0 (absent) or 1 (present) through indicator (dummy) variables. Hence for the occurrence of a particular non-reference level, a correct representation is

(A) 0 1

(B) 0 0 1

(C) 0 1 1

(D) 0 0 0

9. The following matrix **B** positive definite (TRUE/FALSE) **FALSE.**

$$\mathbf{B} = \begin{bmatrix} -1 & 2 \\ 2 & 5 \end{bmatrix}$$

10. Survey Data is collected from different states in India on number of defective electrical vehicles on the road. The data from South zone is not reported due to an unexpected server problem. This type of missing data (given in full form is) is termed as

MISSING COMPLETELY AT RANDOM.

11. A specialized pressure gauge is used to record data from 20 to 40 bar(g) in a reactor. Due to stopping of gas production in the reactor, the instrument stops recording. The missing data (given in full form) may be classified as

MISSING NOT AT RANDOM

12. In Figure 1 given below, the interquartile range is 13.8.

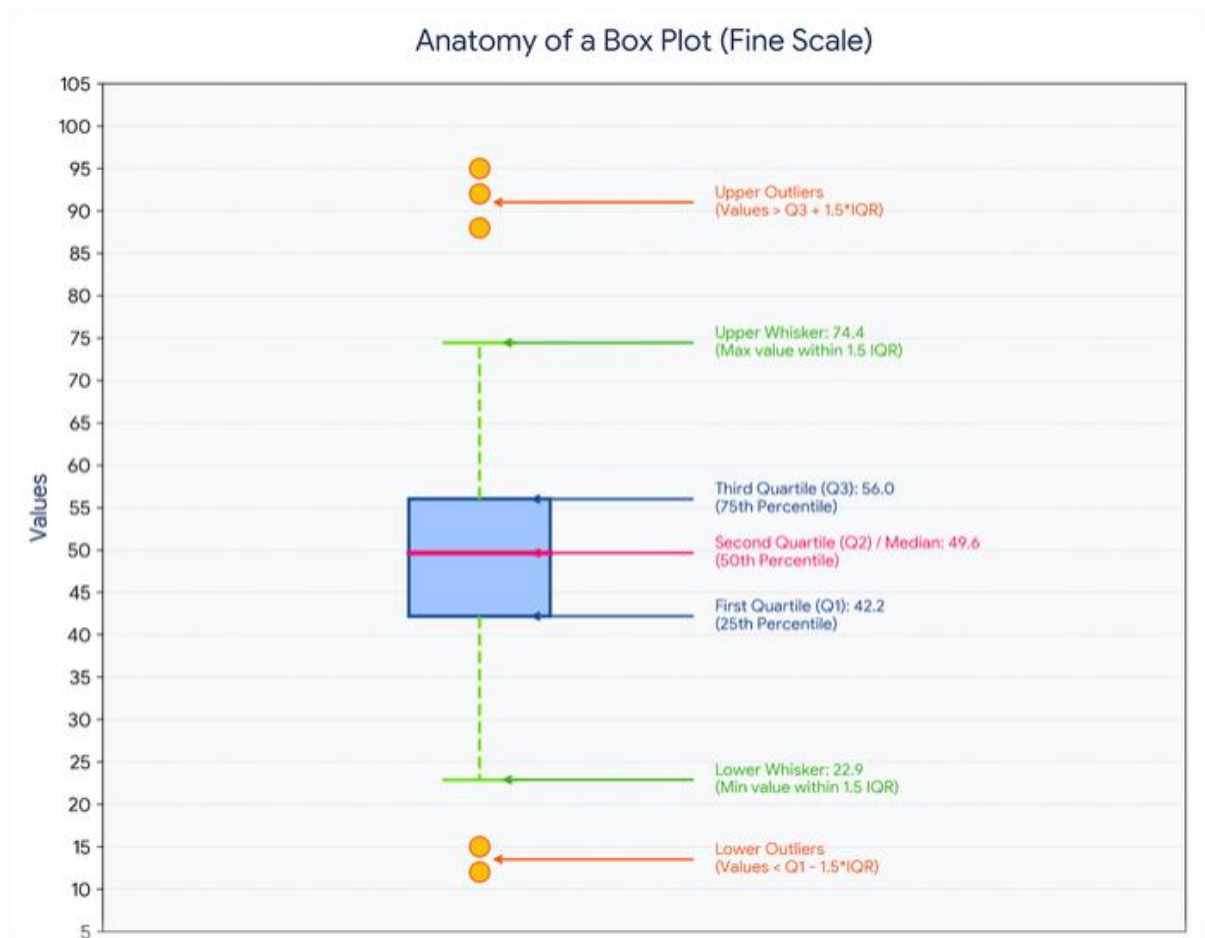


Figure 1. Box plot for Question 12

13. In a column of data corresponding to the age feature, the missing age data are filled by using the mode. This will lead to variance of the data set being

- (A) Reduced
- (B) Remaining same
- (C) Increased
- (D) Increased or decreased depending on the mode value

14. A data column has values of [20, 30, 50, 60]. Find the mean of this data.

40

15. Find the standard deviation (given to two decimal places) of the data set given in Question 14.

18.26

16. Standardize all the numbers of the dataset given in Question 14 and express the answers in a horizontal row below. Separate the standardized values with commas between the numbers i.e., for example a,b,c,d. Give each number to three decimal places.

-1.095, -0.548, 0.548, 1.095

17. In a linear regression problem, there are three features (X_1 , X_2 , and X_3) and two independent responses (Y_1 and Y_2). Let there be 10 observations made at different settings of the features at which the responses Y_1 and Y_2 were measured. The multiple linear regression model includes the intercept β_0 . For this case, the dimensions of the $\mathbf{X}$ matrix are

- (A) 10 x 4
- (B) 4 x 10
- (C) 10 x 3
- (D) 3 x 10

18. For the data given in the above problem (Question 17), the dimensions of $\hat{\beta}$ will be

- (A) 1 x 4
- (B) 4 x 1
- (C) 4 x 2
- (D) 4 x 3

19. When two independent datasets $\mathbf{X}_1$ and $\mathbf{X}_2$ are centered at the origin, the statistical distance and Euclidean distance will be the same when

- (A) Variance of X_1 and Variance of X_2 are both zero
- (B) Variance of X_1 and Variance of X_2 are both exactly unity
- (C) Variance of X_1 and Variance of X_2 are both equal and > 1
- (D) Variance of X_1 and Variance of X_2 are both equal and < 1

Section B

All Questions in this section carry TWO marks each

Let matrix **A** be given as follows

$$\mathbf{A} = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

20. The eigenvalues λ of matrix **A** given below are

- (A) $\lambda_1 = 1$ and $\lambda_2 = -5$
- (B) $\lambda_1 = -1$ and $\lambda_2 = +5$
- (C) $\lambda_1 = -1$ and $\lambda_2 = -5$
- (D) $\lambda_1 = +1$ and $\lambda_2 = +5$

21. Matrix **C** is given as follows

$$\mathbf{C} = \begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

The inverse of the matrix $\mathbf{C}^{-1}$ is given below

- (A) $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$
- (B) $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$
- (C) $\begin{bmatrix} \cos(\theta) & \sin(\theta) \\ -\sin(\theta) & \cos(\theta) \end{bmatrix}$

(D) $\begin{bmatrix} -\cos(\theta) & -\sin(\theta) \\ \sin(\theta) & -\cos(\theta) \end{bmatrix}$

22. Both columns and rows of matrix **C** given above in Question 21 are orthonormal (TRUE/FALSE) and

the matrix **C** itself is orthogonal (TRUE/FALSE)

23. Any idempotent matrix **IMP** satisfies the following relations

$$\mathbf{IMP} = \mathbf{IMP}^2 = \mathbf{IMP}^3 \quad (\text{TRUE/FALSE})$$

$$\mathbf{IMP} \neq \mathbf{IMP}^{0.5} \neq \mathbf{IMP}^{0.25} \quad (\text{TRUE/FALSE})$$

24. In the data series given below, some data are missing and represented by NaN.

$$\mathbf{A} = [12 \quad \text{NaN} \quad 15 \quad \text{NaN} \quad \text{NaN} \quad 20 \quad 22 \quad \text{NaN} \quad 25 \quad 27]$$

Using the window length of 5, and the moving mean method, the 4 missing values are estimated to be (give to two decimal places)

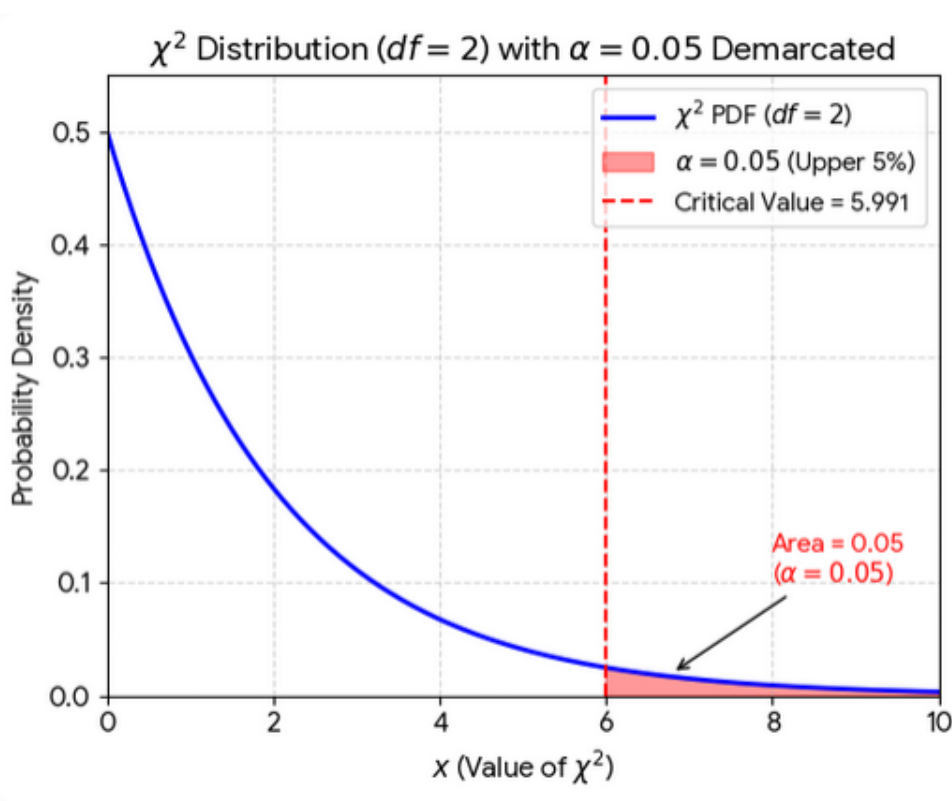
- I. 13.50
- II. 17.50
- III. 19.00
- IV. 23.50

25. Fill in the blanks

Galton's observations may be summarized as follows:

While studying the inheritance of traits, Galton observed that children of very tall parents were usually shorter than their parents but taller than the average, while children of very short parents were usually taller than their parents but shorter than the average.

26. For the Figure 2 plotted below, the maximum statistical distance d from the centre of the ellipse to its boundary (give to two decimal places) is 2.45



To demarcate the area corresponding to $\alpha = 0.05$ for a Chi-Squared (χ^2) distribution with 2 degrees of freedom ($df = 2$), we need to find the critical value where the upper-tail area equals exactly 0.05.

Figure 2. Chi-Square Distribution Plot (Question 26)

Section C: Common Data Problems (Questions 27 – 29)

All Questions in this section carry 3 marks each

Problem Statement

In a linear regression analysis involving two features and one response, the experimental data (both features and responses) was recorded. The resulting data matrix is given below in Table 2.

Table 2. Data for Linear Regression Analysis (Questions 27 – 29)

X_1	X_2	Y
-1	-1	81
-1	1	89
1	-1	83
1	1	91
-1	-1	79
-1	1	87
1	-1	84
1	1	90

27. Find the $(X'X)^{-1}$ matrix. Give to three decimal places for each element of the matrix.

Enter the values of each row of the $(X'X)^{-1}$ matrix separated by a semicolon (;). For example, for a 3x3 matrix **EG** given by

$$\mathbf{EG} = \begin{bmatrix} C11 & C12 & C13 \\ C21 & C22 & C23 \\ C31 & C32 & C33 \end{bmatrix}$$

Is represented as shown below.

$$(X'X)^{-1} = C11,C12,C13;C21,C22,C23;C31,C32,C33$$

$$(\mathbf{X}'\mathbf{X})^{-1} = 0.125, 0, 0; 0, 0.125, 0; 0, 0, 0.125$$

28. Find the $\mathbf{X}'\mathbf{Y}$ matrix. Follow the rule given in Question 27 for representing the answer.

$$\mathbf{X}'\mathbf{Y} = 684, 12, 30$$

29. Find the estimated parameters vector $\hat{\beta}$. Give the transpose of the vector below. Separate your vector components with a comma. Give the numbers to two decimal places

$$\hat{\beta}' = 85.50, 1.50, 3.75$$

30. For the datasets $\mathbf{X}$ and $\mathbf{Y}$ given in Table 3, find the variances and correlation coefficient $\text{corr}(\mathbf{X}, \mathbf{Y})$. Give your answers to three decimal places.

(a) Variance ($\mathbf{X}$) = 3.500

(b) Variance ($\mathbf{Y}$) = 21.500

(c) $\text{Corr}(\mathbf{X}, \mathbf{Y}) = 0.934$

Table 3. Data for finding Covariance (Question 30)

Sl. No.	X	Y
1	1	4
2	2	8
3	3	6
4	4	10
5	5	12
6	6	17

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